General Gunsmithing & Workshop Disciplines

General Workshop • Headspace, Chamber Casting, Machining, Bluing & Diagnostics

Manufacturer:	General Workshop	**Category:**	Workshop Disciplines
Action Type:	Gunsmithing Bench Procedures & Machining Standards	**Era / Service:**	Professional Gunsmithing Standards
Calibers:	Rimmed, Rimless, Belted Magnums, Straight Wall	**Covered Models:**	All Firearms, Bolt Actions, Lever Actions, Semi-Autos, Revolvers

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Headspace (Go / No-Go / Field) • Cerrosafe Chamber Casting • Slow Rust & Hot Caustic Bluing • 11° Target Crowning • Glass Bedding • Sear Stoning

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
HEADSPACE SAFETY RULE	Bolt MUST close on GO gauge; bolt must NOT close on NO-GO with light finger pressure; bolt closing on FIELD is DANGEROUS/UNSAFE
CERROSAFE CASTING WINDOW	Pour alloy at 160°–190°F. Measure cast at EXACTLY 1 hour after pouring for 1:1 true chamber dimension
HOT CAUSTIC BLUING TEMP	285°F to 292°F boiling sodium hydroxide / nitrate bath; exact boiling point indicates salt concentration
TARGET CROWN ANGLE	11° recessed target crown cut perpendicular to bore axis; eliminates uneven muzzle blast gas dispersion
GLASS BEDDING CLEARANCE	0.030" to 0.050" relief around sides and bottom of recoil lug; front of lug takes 100% of recoil bearing

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Diagnostic Protocol 1: Headspace Failure (Bolt Closes on FIELD Gauge) **Bolt handle closes with normal pressure over a hardened steel FIELD headspace gauge**	• Worn receiver locking lugs or barrel set back. • Unsafe bolt swap without gauging headspace.	**Immediate Action: RED TAG FIREARM: UNSAFE TO FIRE. Case head separation risk. Gunsmith Action: Set barrel back by cutting 1 full thread pitch on lathe, recut chamber with finish reamer to exact GO depth, or install oversized locking block/bolt head.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Headspace Gauging Masterclass: GO, NO-GO & FIELD Explained	Larry Potterfield / MidwayUSA	Master gunsmith Larry Potterfield demonstrates how headspace is measured across rimmed, rimless, and belted cases, and how to verify bolt safety.